

Poojitha Panabaka

Software Engineer

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years of experience building **scalable distributed systems, cloud-native microservices**, and high-performance **full-stack applications** using **Java, Spring Boot, Angular, React, Python, and AWS**. Strong in **backend engineering, event-driven architectures, asynchronous processing**, and **API design**, with experience in **CI/CD automation** and **cloud infrastructure at scale**. Focused on **system performance optimization, observability, reliability engineering**, and **production-grade distributed systems**, including **compliance-driven** and **AI-assisted engineering workflows**.

SKILLS

Programming Languages: Java, Python, TypeScript, JavaScript, Go, SQL

Backend Development: Spring Boot, Spring MVC, REST APIs, Microservices, Distributed Systems, GraphQL, Kafka, RxJava, Node.js, OAuth2, JWT, API Gateway, Redis, Event-Driven Architecture, Concurrency, Design Patterns

Frontend Development: Angular, React, RxJS, TypeScript, JavaScript, HTML, CSS, Bootstrap, Material UI, Responsive UI Design, Component-Based Architecture, State Management

Cloud & DevOps: AWS (S3, ECS, DynamoDB, CloudWatch, IAM, CodePipeline, CodeDeploy, Lambda, API Gateway), Docker, Kubernetes, Jenkins, GitHub Actions, Terraform, CI/CD Automation, SonarQube, Observability, Monitoring, Logging

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Firebase

Testing & Quality Engineering: JUnit, PyTest, Selenium, Jasmine, Karma, Integration Testing, Regression Testing, Test Automation, Static Code Analysis

AI-Assisted Engineering: LLM Workflow Automation, MCP Integrations, AI Developer Tooling, Context-Aware Engineering Workflows, Developer Productivity Automation

Engineering Practices: System Design, Distributed Systems, Scalability, Reliability Engineering, Performance Optimization, Agile/Scrum, Git, Bitbucket, JIRA, Code Reviews

PROFESSIONAL EXPERIENCE

Moody's Corporation | Associate Software Engineer

Sep 2023 - Present

- Eliminated persistent **OOM failures** during **200MB+ document uploads** by migrating from in-memory buffering to **Streaming I/O using AWS S3 Transfer Manager**, reducing upload latency by **50% (10 min → 4 min)** and enabling scalable large-file processing.
- Built **Kafka-driven asynchronous processing pipelines** for post-upload document workflows, enabling scalable **event-driven architecture** between document ingestion and downstream **EDMS processing services**, while reducing coupling across distributed systems.
- Architected a **Strategy Design Pattern** in **Java Spring Boot** to decouple consumer-specific workflows from core document processing, scaling the platform to support **20+ integrations** without modifying core business logic.
- Improved **Angular platform responsiveness** by **30%** and reduced **page load times** by **20%** for **10K+ daily users** by optimizing change detection and replacing resolver-based navigation with **reactive RxJS workflows**.
- Designed **RBAC-based access control system** and **AWS S3 lifecycle tiering (Standard → Glacier)** for compliance-driven retention across **millions of financial records**, ensuring regulatory audit readiness while optimizing storage costs.
- Identified and resolved silent **EDMS/PostgreSQL data integrity drift** during a file-name backfill migration by cross-referencing distributed data sources, uncovering a **delete-propagation defect**, and establishing a **multi-phase deployment strategy** to prevent recurrence.
- Introduced a **PostgreSQL-backed Change History audit system** capturing before/after deletion states, enabling **compliance-grade traceability** for enterprise document management workflows.
- Built a **Python-based API automation framework** covering end-to-end authentication and service contract validation; integrated with **GitHub Actions CI/CD**, reducing manual QA effort by **35%**.
- Engineered **Jenkins CI/CD pipelines** with **SonarQube quality gates** and automated release validation, improving deployment reliability and reducing release bottlenecks across multi-environment deployments.
- Integrated in-house **LLM tooling with MCP servers (Git + JIRA automation)** to automate branch creation, code scaffolding, and review workflows, increasing engineering delivery velocity by **5x**.
- Facilitated cross-functional planning between engineering, product, and compliance teams; introduced **dependency-tracking workflows** that accelerated feature delivery by **30%** and reduced compliance review cycles from **3 weeks → <1 week**.

Tech Stack: Java, Spring Boot, Angular, RxJS, Python, PostgreSQL, AWS S3, ECS, DynamoDB, Jenkins, GitHub Actions, SonarQube, Docker, REST APIs, Distributed Systems, CI/CD Automation, Observability, AI Developer Tooling

Tata Consultancy Services | Assistant System Engineer

Sep 2020 - Dec 2021

- Developed scalable **Python-based microservices and backend APIs** for Walmart enterprise systems, improving **distributed system workflow automation**, backend reliability, and service processing efficiency.
- Containerized and orchestrated applications using **Docker and Kubernetes**, improving **deployment consistency** and

streamlining **multi-environment CI/CD release workflows**.

- Built responsive **React + TypeScript frontend applications** with **Bootstrap**, delivering cross-device compatible user experiences for enterprise client systems.
- Implemented **automated testing pipelines (PyTest, Selenium)** achieving **85%+ code coverage**, reducing **production defects by 30%** and improving release stability.
- Collaborated within **Agile teams** to deliver backend services, frontend features, and **CI/CD pipeline improvements** across enterprise retail systems.

Tech Stack: Python, React, TypeScript, Docker, Kubernetes, REST APIs, PyTest, Selenium, CI/CD, Distributed Systems, Agile Development

Codinza | Full-Stack Developer Intern

Mar 2019 - Jul 2019

- Built and deployed **responsive single-page applications (SPAs)** using **React, TypeScript, and Vue.js**, improving **frontend performance, responsiveness, and user engagement**.
- Integrated **REST APIs** with **Angular-based frontend workflows**, validating **end-to-end CRUD operations** built on **Spring Boot backend services** to ensure reliable data flow across full-stack systems.
- Translated **UI/UX wireframes into production-ready responsive interfaces**, ensuring **cross-browser compatibility**, consistent user experience, and adherence to modern web standards.

Tech Stack: React, Angular, Vue.js, TypeScript, Spring Boot, REST APIs, HTML, CSS, JavaScript

PROJECTS

QourtPlay - Sports Scoring & Ranking Platform — [App Store](#)

- Developed **backend services for a real-time sports scoring and ELO-based ranking platform** using **Python, Docker, and AWS ECS**, supporting **low-latency live match processing and competitive scoring workflows**.
- Implemented **Keycloak-based authentication and authorization system**, securing access across **mobile and web clients** with role-based access control and centralized identity management.
- Built **NFC-enabled IoT check-in system (PN532 modules)** to automate **player attendance tracking and match lifecycle events**, improving operational efficiency and reducing manual logging.
- Supported **production deployments, QA validation, and App Store release cycles**, ensuring reliability and stability of a **live consumer-facing platform**.

Tech Stack: Python, AWS ECS, Docker, Keycloak, NFC, IoT, REST APIs

EDUCATION

University of North Carolina at Charlotte

Master of Science in Information Technology | May 2023

Relevant Coursework: Algorithms & Data Structures, Network-Based Application Development, Mobile App Development, Database Systems, Visual Analytics